

UNIT-2

Datacenter Overview, Data center evolution, Modern business requirements for data center, making agile datacenter, data center transformations, future of data centers

DataCenter Overview

The modern data center is a rapidly evolving environment that looks nothing like the data centers of a decade ago. It is heavily influenced by Moore's Law, which states that computing power roughly doubles every 2.5 to 3 years. Because of this exponential growth, tasks that were previously considered cost-prohibitive or physically impossible—such as using software defined storage (SDS) to serve I/O to Tier 1 applications—are now commonplace.

1. IT Is Changing... and It Must

As bleeding-edge technology trickles into mainstream IT, business leaders must dramatically transform how they operate. Those who embrace these new developments will thrive, and those who do not will struggle to survive. The exponential growth in technology is generating entirely new industries, drastically changing everyday experiences from shopping to transportation.

2. Simplification Is the New Black

Thanks to the Internet of Things (IoT), the scope of IT's responsibility is constantly expanding. A 2014 Gartner report estimated that there would be 25 trillion connected devices by 2020. This includes sensors, medical devices, lights, and actuators exchanging data on private and public networks.

Because IT is tasked with managing these endpoints, the primary goal for future IT departments is to radically simplify system administration. Manufacturers must design products that can self-configure, self-heal, and make intelligent decisions to reduce administrative burden. IT Directors and CIOs must ruthlessly eliminate organizational complexity and silos to replace productivity-sapping tasks with elegant simplicity.

3. Focus on Cost Reduction

Despite the exponential increase in managed data and endpoints, a Gartner report predicted that IT budgets would remain essentially flat through 2020, even while 55% of respondents expected their overall revenues to increase. Therefore, IT departments must find ways to complete large projects for a fraction of the historical cost. Fortunately, due to denser processors, cheaper RAM, and falling solid-state drive (SSD) costs, highly complex projects can now be executed within constrained budgets.

4. Focus on Customer Service and the Business

IT professionals no longer have the luxury of spending days tinkering with solutions. They expect "white-glove" support from vendors. More importantly, IT must focus on delivering exceptional value to internal business customers. If IT fails to provide agility and high-quality services, business units will bypass them, giving rise to "Shadow IT" (e.g., using personal Dropbox accounts instead of

corporate SharePoint). Shadow IT creates massive security, compliance, and control risks for the entire business.

5. Adapt or Die & The Flattening of the IT Organization

In previous eras, data centers were strictly siloed, containing specialized administrators for storage, networks, and servers. Today, these silos are dissolving. The mainstream adoption of x86 virtualization brought these technologies closer together, forcing IT professionals to become highly proficient generalists. IT professionals must adapt and expand their skillsets across all domains, or risk becoming obsolete.

6. IT as an Operational Expense (OpEx)

Traditional data centers functioned on Capital Expenditures (CapEx), where organizations bought hardware upfront based on multi-year projections, leading to massive immediate depreciation. Modern IT budgets heavily favor Operational Expenditures (OpEx). Cloud resources and Software-as-a-Service (SaaS) platforms allow organizations to pay monthly based on exact consumption. SDS and hyperconverged infrastructures bring these pay-as-you-grow, OpEx-style economic models directly into the on-premises data center.

Data Center Evolution

The data center has historically been cyclical, but every iteration improves substantially. The foundation of the modern data center rests on two continuously developing technologies: the microprocessor (x86 architecture) and disk-based storage media.

1. The Rise of the Monolithic Storage Array

In the 1990s, the prevailing design was direct-attached storage (DAS) localized to each individual server. This caused two massive inefficiencies:

1. **Stranded Compute:** Servers regularly ran at 10% CPU utilization, wasting massive computational potential.
2. **Stranded Capacity:** If an enterprise had 500 servers and left 50 GB of unused storage on each to allow for growth, 25 TB of expensive storage sat completely idle.

To solve this, disks were pooled onto a network segregated from the local area network (LAN), called a Storage Area Network (SAN). The SAN utilized Fibre Channel Protocol to ensure high-speed, lossless data transmission. This allowed multiple servers to access a shared storage array, increasing utilization and unlocking advanced data services like volume-level snapshots, replication, and global data deduplication.

2. The Virtualization of Compute — Software Defined Servers

To fix stranded compute resources, VMware released ESX in 2001, a bare-metal hypervisor that allowed multiple virtual workloads to run simultaneously and isolated on shared hardware. Prior to this, a 2006 EPA study estimated that data centers consumed 61 billion kilowatt-hours of electricity (1.5% of total U.S. consumption, costing roughly \$4.5 billion).

Server virtualization allowed for Physical-to-Virtual (P2V) migrations. This yielded massive consolidation ratios (e.g., condensing 4 physical servers into 4 virtual machines on 1 host), heavily reducing physical footprints, cooling requirements, and power consumption.

3. The No-Spin Zone: The Move from Disk to Flash

The physical limitations of magnetic spinning disks became a severe bottleneck. Rotational speeds maximized at 15,000 RPM (15K RPM) around the year 2000, limiting performance to roughly 150-200 Input/Output Operations Per Second (IOPS). The non-mechanical nature of Flash storage (solid-state drives) solves this. Flash delivers microsecond latency and exponentially higher IOPS, while consuming far less power. Modern Enterprise Multi-Level Cell (eMLC) flash drives are highly durable, commonly warrantied for 10 full drive writes per day over 5 years.

4. The Fall of the Monolithic Storage Array

Monolithic storage arrays failed to keep up with the demands of highly dense virtualization. Virtual Desktop Infrastructure (VDI) and server workloads occurring simultaneously on the same array created the "I/O Blender" effect—highly randomized I/O operations that crippled legacy arrays. This drove the industry back toward direct-attached storage, but with a software-defined twist.

5. The Emergence of Convergence & Cloud

Convergence evolved to combat the complexity of assembling infrastructure. A converged infrastructure arrives pre-cabled and pre-configured based on tested reference architectures, combining compute, storage, and networking into a single rack.

Cloud computing took this agility further by delivering resources flexibly, rapidly, and on-demand.

- **Public Cloud:** Resources provisioned in a third-party data center (e.g., AWS, Azure).
- **Private Cloud:** On-premises deployments providing the same rapid, on-demand automated provisioning.
- **Hybrid Cloud:** Combining both to allow "bursting" workloads to the public cloud during high-demand periods (like retail Black Friday traffic) while keeping sensitive databases locally.

Modern Business Requirements for Data Center

Infrastructure exists to run applications that meet the business's needs, but business requirements are notoriously dynamic. Modern designs must solve five core challenges: Risk, Cost, Complexity, Agility and Performance, and Resiliency and Availability.

1. Risk

In IT, risk translates to the potential loss of the company's data, its most valuable asset. The data center must balance the CIA triad:

- **Confidentiality:** Handled via authentication, authorization, and accounting (AAA) to ensure privacy.
- **Integrity:** Ensuring data hasn't been modified by unauthorized individuals, maintained via AAA and encryption.
- **Availability:** Ensuring data is accessible to end users and applications whenever needed.

IT must also ensure Business Continuity (BC) and Disaster Recovery (DR) by performing risk assessments and securing top management financial commitment. Furthermore, IT faces regulatory risks (like SOX for publicly traded companies, PCI for credit cards, and HIPAA for medical records) and technical risks like vendor lock-in (hypervisor, storage, or server constraints). IT must overcome its legacy perception as the "NO" department by positioning itself as a strategic business-enabler.

2. Cost

Businesses exist to generate profit. IT must shift away from the traditional 3-to-5-year CapEx purchasing model, which involves massive upfront costs and immediate depreciation. The business requires OpEx models that allow for granular, pay-as-you-go scaling. This budgetary shift is changing the IT career landscape. Administrators must elevate their skill sets from siloed tasks (like provisioning LUNs) to overarching infrastructure architect roles.

3. Complexity

There is zero business requirement for complexity. Complexity acts only as a hurdle that necessitates expensive specialized technologists. Modern Software Defined Data Center (SDDC) solutions eliminate the complexity born from hardware-defined legacy systems, allowing the business to manage vast infrastructures through simplified, unified interfaces.

4. Agility and Performance

Agility is a company's ability to adapt swiftly to changing competitive environments. This translates to deploying and scaling applications almost instantly.

- **Automation:** Scripting common, individual tasks (e.g., deploying a new Windows server via a single command).
- **Orchestration:** Linking automated tasks to complete complex business processes (e.g., deploying an entire 3-tier web, middleware, and database application environment).

This leads to **Self-Service**, allowing developers and power users to dispense infrastructure exactly when they need it via a service catalog, ensuring the IT department maintains governance without acting as a bottleneck.

5. The Data Growth Challenge & Resiliency

According to the United Nations Economic Council, global data will more than double to over 40 Zettabytes by 2020. Oracle research indicates that by 2020, IT departments will manage 10 times the servers and 50 times the data with only 1.5 times the staff. To meet this challenge, IT requires advanced data reduction services (deduplication and compression) inherent in SDS. End-users, accustomed to the uninterrupted availability of SaaS apps (like Gmail or Facebook), now demand zero downtime from internal enterprise applications, necessitating highly resilient, distributed software-defined architectures.

Making Agile Data Center

To build an agile IT infrastructure, data center managers must follow three key principles: Think Big, Start Small, and Move Fast.

1. Think Big

Organizations must re-evaluate everything. Agile IT demands discarding old ideas about hardware procurement and support. For example, if your IT staff is providing break/fix services for desktop computers, they are incurring an "opportunity cost." For every commodity service kept in-house, less time is devoted to value-add projects that generate business revenue. IT must eliminate complex hardware layers and implement automation and orchestration broadly to ensure the underlying infrastructure is bereft of complexity.

2. Start Small

Total rip-and-replace methodologies are rarely feasible. Find a specific problem and fix it. If the goal is to move the entire data center to a hyperconverged infrastructure (HCI), use an upcoming Virtual Desktop Infrastructure (VDI) project as the initial stepping stone. Staging implementation minimizes production impact. Start small with automation as well: write a script to automate user account provisioning, turning the administrator's role from manual execution to automated oversight.

3. Move Fast (Six Implementation Steps)

1. **Talk to the Business:** Interview business unit leaders to identify objectives and technology support gaps before speaking with the IT department.
2. **Assess the Technology Environment:** Evaluate if the existing, highly customized hardware morass can actually support the newly identified business needs. Create a granular list of all products and the business need they serve.
3. **Create a Support Inventory:** Analyze internal IT processes. Are there staff skill deficiencies or a lack of business focus preventing IT from executing strategic priorities?
4. **Identify Core Services and Gaps:** Determine exactly which products are core to the business's mission and where the IT support structure falls short.
5. **Decide What to Outsource:** Outsource commodity services to make them "someone else's problem." For example, outsource Microsoft Exchange to Office 365, allowing staff to redirect efforts to value-add projects.
6. **Decide What to Improve:** Assign two metrics (from 1 to 5) to every IT challenge: one for *Difficulty* and one for *Payoff*. Immediately attack the items with low difficulty (1 or 2) and high payoff (4 or 5) to generate rapid momentum.

Data Center Transformations

Data center transformation is only valuable if it positively impacts the bottom line. Jack Welch noted that rapid action on gained knowledge is the ultimate competitive advantage. Once objectives are aligned with business needs (e.g., lowering operational expenses, increasing product accuracy, accelerating developer time-to-market), organizations should aggressively target the low-hanging fruit.

1. Where to Address the Low-Hanging Fruit

- **Test/Dev:** Environments are spun up and destroyed constantly. SDS allows rapid copying and deduplication of production data for testing. SDN provides sandboxing to prevent developer environments from crossing wires with production networks.
- **Remote/Branch Offices (ROBO):** Supporting distributed sites is difficult. (A survey notes that 19% of enterprises support 25 or more ROBO sites). Software Defined Compute (SDC) and SDS eliminate the need for dedicated, large-footprint Tier 2 data centers and specialized IT staff at branch offices.
- **Server Virtualization:** By adding configuration management tools like Puppet or Chef to existing virtualization, administrators can orchestrate massive provisioning tasks seamlessly.
- **Big Data/Analytics:** SDC can dynamically create and destroy job-processing nodes on the fly to eliminate idle resources, while SDS scales capacity effortlessly to handle massive analytical datasets.

- **Disaster Recovery (DR):** SDS enables policy-based, highly granular replication. SDN can automatically recreate network topologies at the recovery site, orchestrating seamless failovers.
- **Virtual Desktop Infrastructure (VDI):** VDI breaks the cycle of hardware desktop refreshes. SDS solves the notorious performance bottleneck of VDI "boot storms" through flash-accelerated hyperconvergence.

2. How to Adopt SDDC

Adopting the Software Defined Data Center requires viewing operations strictly through an SDDC lens.

- **Software Defined Compute:** Organizations must choose between Cloud vs. On-Premises, and Virtual Machines vs. OS-level Containers. Implementing a cloud computing management suite on top of the hypervisor unlocks true self-service multi-tenancy.
- **Software Defined Storage:** Organizations must define their provisioning models (monolithic vs. hyperconverged) and storage media (flash vs. disk). SDS decouples the storage intelligence from the physical hardware, allowing policy-based management, deduplication, thin provisioning, and data mobility.
- **Software Defined Networking:** SDN decouples the control plane from the data plane. It allows for advanced, programmatic network scaling and the implementation of micro-segmentation, securing east-west traffic within the data center without requiring expensive physical firewalls.

3. Simplification and Management

Complexity is the costliest attribute of the data center. IT must focus on simplification by eliminating monolithic storage arrays. The best approach is "opportunistic hyperconvergence"—deploying HCI for specific new projects (like VDI or new remote offices) rather than an immediate rip-and-replace. Finally, organizations must implement full-stack management tools that provide holistic visibility across compute, storage, and networking, ultimately enabling the holy trinity of the modern data center: Automation, Orchestration, and Self-Provisioning.

Future of Data Centers

Moore's Law ensures the data center will look drastically different in 3 to 4 years. Technologies currently in development will fundamentally change how computing power and storage are distributed and consumed.

1. Containers Instead of Virtual Machines

While virtual machines (VMs) abstract the operating system from the underlying hardware, containers (like LXC and Docker) abstract the application from the operating system. Containers share the host OS kernel and binaries/libraries, making them incredibly lightweight and efficient compared to VMs.

Historically, vendors shipped software as an Open Virtualization Archive (OVA) file containing a full, pre-configured VM. Moving forward, vendors will ship pre-packaged applications as containers. Future developments aim to ensure cross-platform container support, allowing a container to seamlessly run on either a Linux or Windows host.

2. Open Source Tools

Web giants like Google and Facebook have pioneered massive open-source projects. Due to collaborative global development, open source software often evolves faster and with higher quality than proprietary alternatives. Furthermore, open-source projects are driven by user-base priorities rather than vendor lock-in constraints. In the future, enterprises will default to open-source software, using proprietary solutions only as a last resort.

3. Flash Capacity and 3D NAND

Flash capacity has historically been limited. Traditionally, flash storage density was increased by shrinking the physical size of planar (two-dimensional) cells. Hitting physical limitations, manufacturers like Intel and Micron developed **3D NAND** (e.g., 3D XPoint).

By stacking flash cells vertically, manufacturers can utilize larger, more stable cells while exponentially increasing drive capacity. Consumer-ready 15 TB SSDs are already emerging, which will effectively render magnetic spinning disks obsolete in the near future.

4. Non-Volatile Memory Express (NVMe)

Modern flash is too fast for legacy buses. The Advanced Host Controller Interface (AHCI) standard was built for mechanical drives, featuring a single command queue with a depth of 32 commands. **NVMe**, utilizing the PCIe bus, offers 65,535 individual queues, each with a depth of 65,535 commands. Furthermore, via the MSI-X extension, NVMe allows

up to 2,048 interrupts. This massive parallelism effectively doubles IOPS performance at certain queue depths.

5. Beyond Today's Flash

As NAND flash eventually reaches its absolute physical limit, new non-volatile memories will take over:

- **Phase Change Memory (PCM):** Uses chalcogenide glass (the same material in fiber optics). It changes the atomic state of the material to represent binary data, offering incredible endurance.
- **Resistive RAM (RRAM/Memristor):** Changes the electrical resistance of a dielectric material. It features high durability, high performance, and ultra-low power consumption.
- **Magnetoresistive RAM (MRAM):** Uses magnetic charges instead of electrical charges to measure resistance.

6. Pooling of Resources (Rack Scale Architecture)

Intel is developing Rack Scale Architecture, which completely disaggregates the traditional server. Instead of individual servers in a rack, the entire rack becomes the resource pool.

High-speed silicon photonics fabrics connect specialized trays: a tray of pure CPUs, a tray of pure RAM, and a tray of pure NVMe storage at the bottom. This allows IT to upgrade just the CPUs without having to throw away motherboards, memory, and network cards, drastically shifting upgrade economics.

7. The Cloud

Data centers will heavily adopt Hybrid Cloud architectures. The future will blur the lines between public and private clouds, utilizing abstraction tools to manage multiple distinct environments as one cohesive resource pool.

- **Workload Mobility:** Applications will seamlessly migrate between on-premises and public clouds based on real-time electricity/resource costs or seasonal traffic demands.
- **Cloud Brokering:** Automated workload placement engines will programmatically determine which cloud provider among multiple options offers the best environment for a specific app deployment.
- **Security:** Cloud security will rely heavily on end-user competence. Gartner speculates that by 2020, 95% of cloud security incidents will be the fault of the customer (misconfiguration) rather than the cloud provider. Education and strict cloud-accounting tools will be the primary defenses for the future data center.
